

Computer Networks I

Data Link Control Protocols (Error Control)

Amitangshu Pal
Computer Science and Engineering
IIT Kanpur

Error Control

❑ Two types of error

- **Lost frame:** A frame fails to arrive at the other end
- **Damaged frame:** A frame arrives at the destination, but few bits are damaged

❑ Automatic Repeat Request (ARQ):

- Stop-and-wait ARQ
 - Go-back-N ARQ
 - Selective-reject ARQ
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Stop-and-Wait ARQ

Stop-and-Wait ARQ

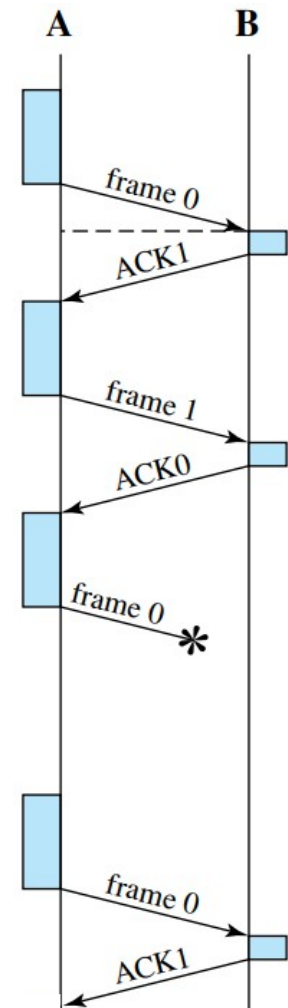
❑ Based on Stop-and-Wait flow control

❑ Source:

- Source transmits single frame, waits for an ACK
- It also starts a timer and maintains the copy
- If ACK received, stop timer and transmit next frame
- Sender has timeout → if no ACK within timeout, retransmit

❑ Destination:

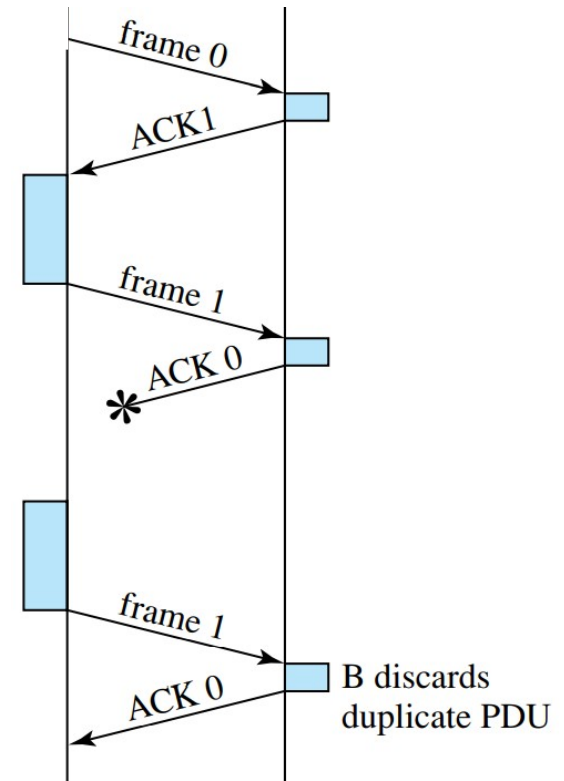
- Destination sends ACK if frame received correctly (with no errors)
- If frame received is damaged, discard it



Stop-and-Wait ARQ

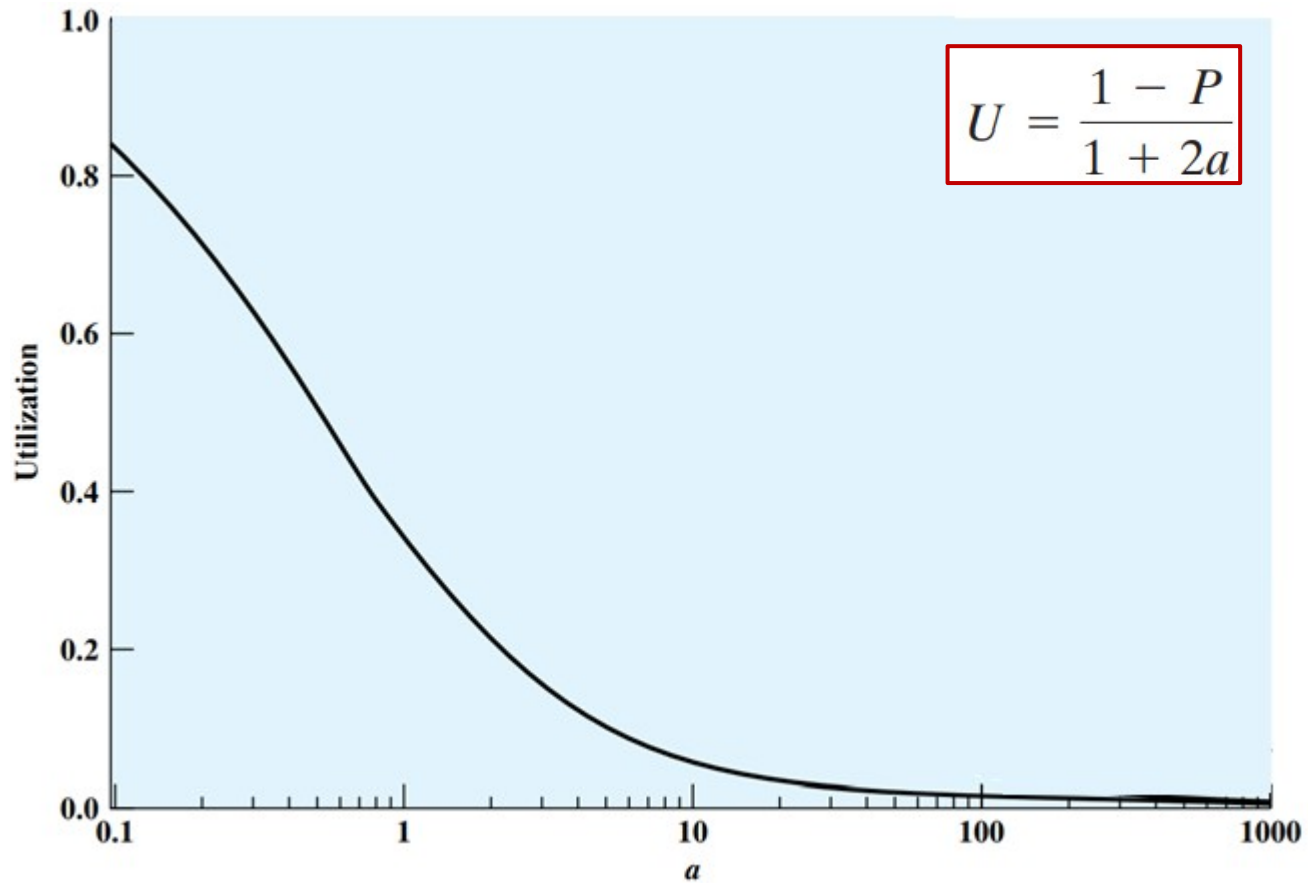
- ❑ If ACK is damaged, transmitter will not recognize
 - Transmitter will retransmit after the timeout
 - Receiver gets two copies of frame
 - **Solution:** use 1-bit frame SEQ number and ACK0 / ACK1

- ❑ Stop-and-Wait ARQ is **simple, but inefficient**



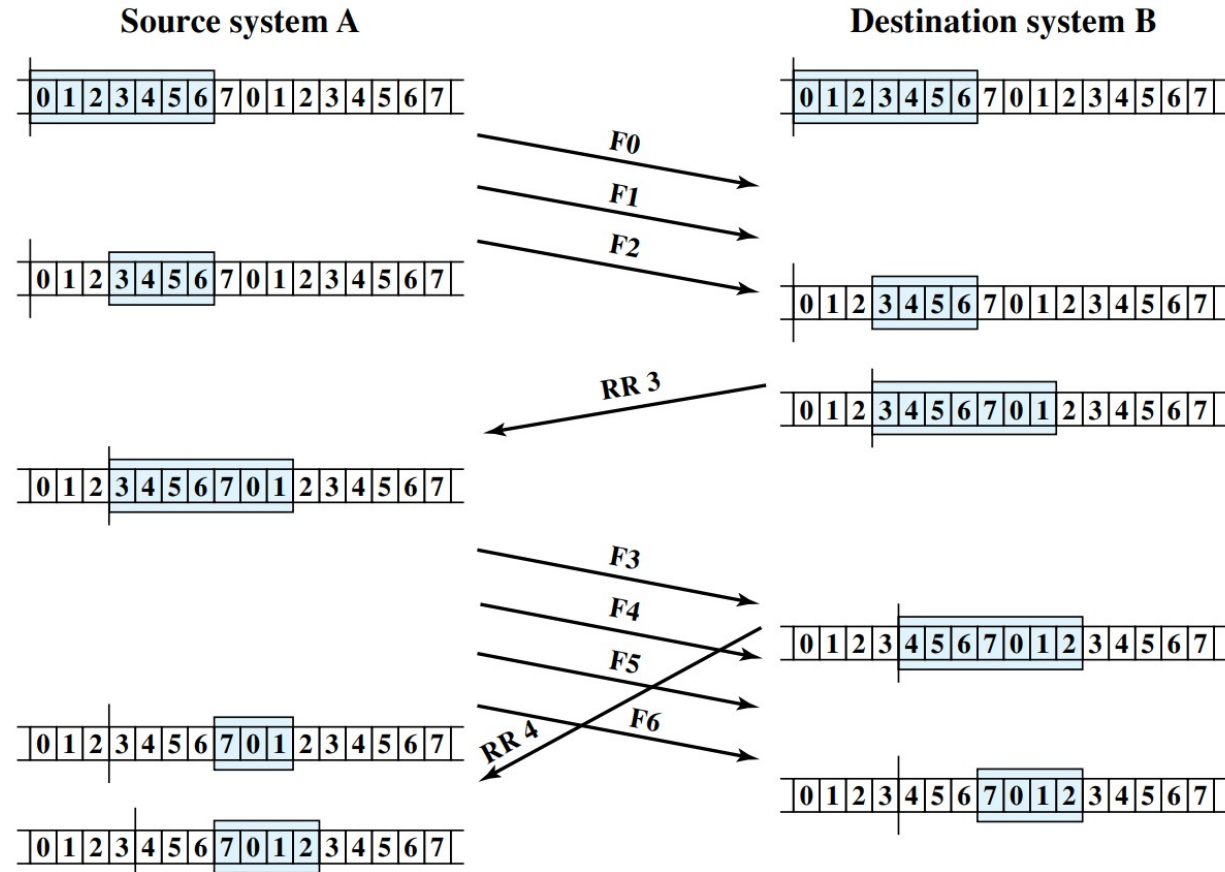
Stop-and-Wait ARQ

$$a = \frac{t_{\text{prop}}}{t_{\text{frame}}}$$



Go-Back-N ARQ

Sliding Window Flow Control: Example



Go-Back-N ARQ

- ❑ Based on Sliding Window flow control → Most commonly used error control

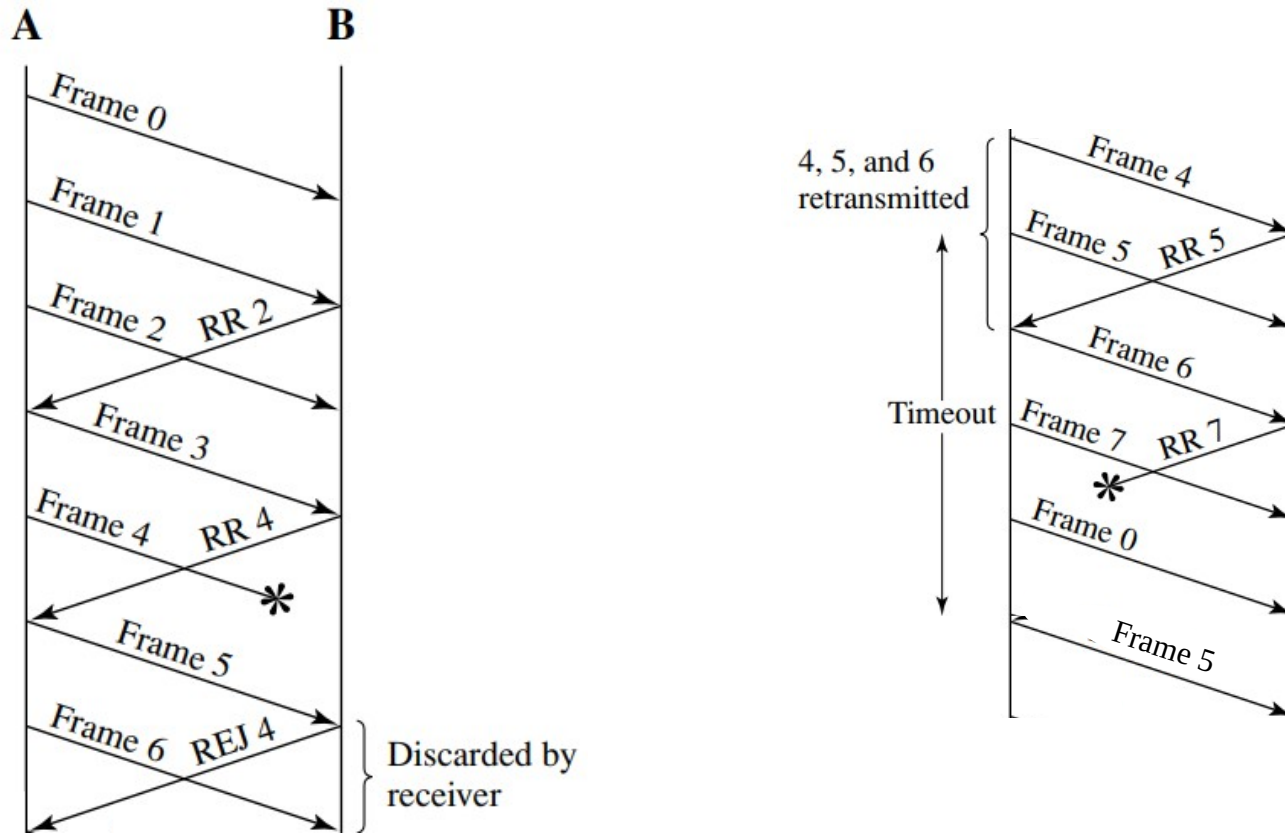
❑ Source:

- Sender must go back and retransmit that frame and all subsequent frames
- If no ACK received from Destination after timeout, the Source can retransmit the previous frame asked by the receiver

❑ Destination:

- If no error, the destination will ACK incoming frames as in sliding window
 - RR=receive ready, or piggybacked acknowledgment with sequence number of next expected frame
 - If the destination station detects an error in a frame, it may send a negative acknowledgment
 - REJ=reject with sequence number of next expected frame
 - Destination will discard that frame and all future frames until the frame in error is received correctly
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Go-Back-N ARQ



Go-Back-N ARQ

- ❑ A k -bit sequence number provides a sequence number range of 2^k
 - However the maximum window size is limited to $2^k - 1$
 - ❑ As a case of 3-bit sequence number (i.e. sequence number space is 8)
 - ❑ Suppose sender sends frame 0 and gets back an RR 1
 - ❑ Then sends frames 1, 2, 3, 4, 5, 6, 7, 0 and gets another RR 1
 - ❑ This could mean that all eight frames were received correctly and the RR 1 is a cumulative acknowledgment
 - ❑ It could also mean that all eight frames were damaged or lost in transit, and the receiving station is repeating its previous RR 1
 - ❑ The problem is avoided if the maximum window size is limited to 7, i.e. $2^3 - 1$
 - ❑ Max window size = $2^k - 1$ (for a k -bit sequence number)
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Selective Reject ARQ

Selective Reject ARQ

- ❑ Also called selective retransmission
- ❑ Only rejected or timeout frames are retransmitted
- ❑ Subsequent frames are accepted by the receiver and buffered

❑ Pros:

- Minimizes retransmission

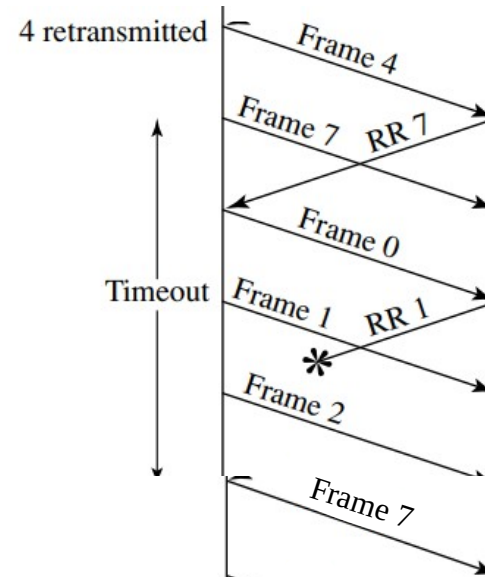
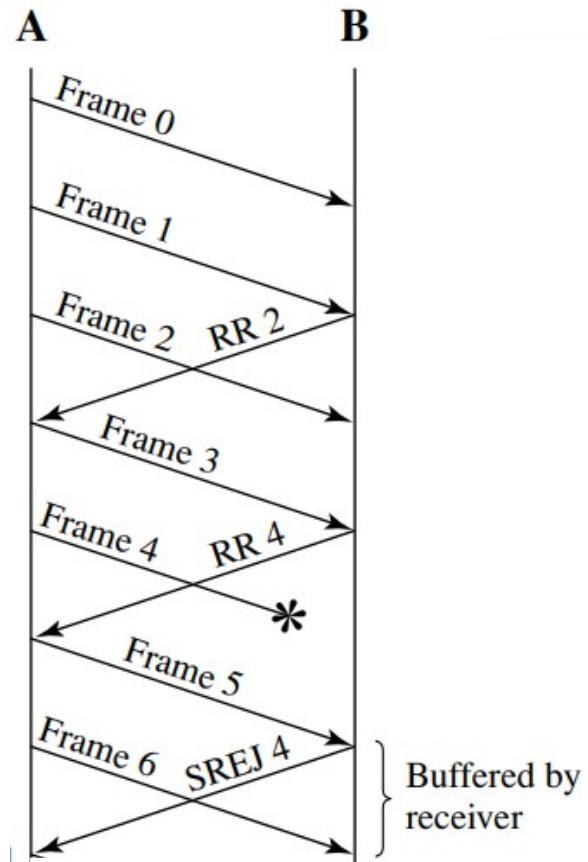
❑ Cons:

- Receiver must maintain large enough buffer to store out-of-order frames
- More complex logic in transmitter

❑ Less widely used

❑ Useful for satellite links with long propagation delays

Selective Reject ARQ



Selective Reject ARQ

- ❑ Window size is more restrictive for selective-reject than for go-back-N
 - ❑ Consider the case of a 3-bit sequence number size for selective-reject
 - Sender sends frames 0 through 6
 - Receiver receives all seven frames and cumulatively acknowledges with RR 7
 - Because of a noise burst, the RR 7 is lost
 - Sender times out and retransmits frame 0
 - Receiver has already advanced its receive window to accept frames 7, 0, 1, 2, 3, 4, and 5
 - It assumes that frame 7 has been lost and that this is a new frame 0, which it accepts
 - ❑ The problem is that there is an overlap in between the sending and receiving window
 - To overcome the problem, the maximum window size should be **no more than half the range of sequence numbers**
 - ❑ Max window size = 2^{k-1} (for a k-bit sequence number)
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Comparison

Flow control

$$U = \frac{1}{1 + 2a}$$

**Stop and
Wait**

$$U = \begin{cases} 1 & W \geq 2a + 1 \\ \frac{W}{2a + 1} & W < 2a + 1 \end{cases}$$

**Sliding
Window**

Error control

$$U = \frac{1 - P}{1 + 2a}$$

**Stop and
Wait**

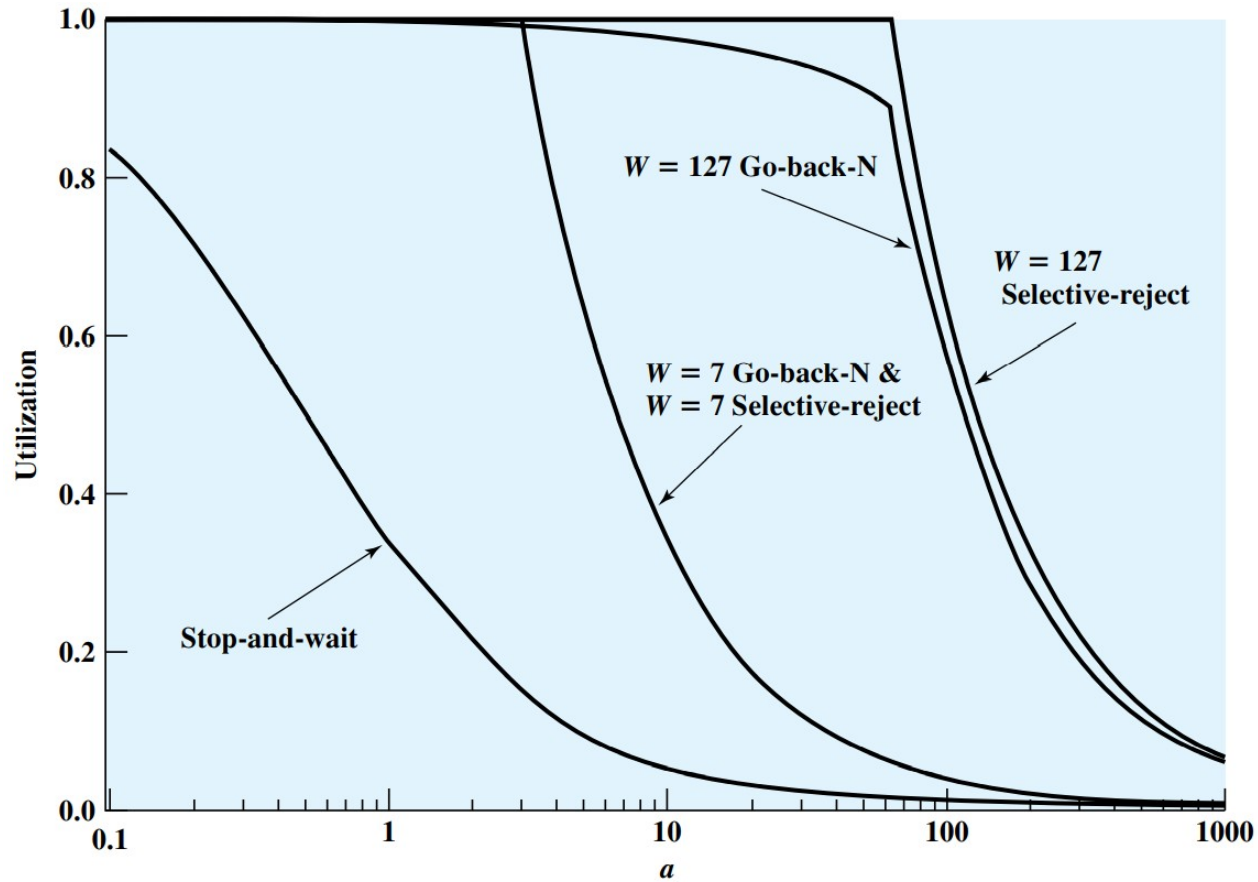
$$U = \begin{cases} \frac{1 - P}{1 + 2a} & W \geq 2a + 1 \\ \frac{W(1 - P)}{2a + 1} & W < 2a + 1 \end{cases}$$

**Selective
Reject**

$$U = \begin{cases} \frac{1 - P}{1 + 2aP} & W \geq 2a + 1 \\ \frac{W(1 - P)}{(2a + 1)(1 - P + WP)} & W < 2a + 1 \end{cases}$$

Go-back N

Comparison



Summary

❑ Error control in Link Layer:

- Stop and wait
 - Selective reject
 - Go-back N
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